

Markscheme

Mathematics: analysis and approaches
Practice question bank

115. (a)

Consider the expansion of $(1+x)^{-1}(1+2x)^{\frac{1}{2}}$ in ascending powers of x , up to the term in x^2 .

Find the coefficient of x^2 .

$$(1+x)^{-1} = 1 - x + x^2 - \dots \text{ and } (1+2x)^{\frac{1}{2}} = 1 + x - \frac{1}{2}x^2 + \dots \quad (M1) \text{ A1}$$

$$\text{multiply and collect the } x^2 \text{ terms: } 1\left(-\frac{1}{2}\right) + (-1)(1) + (1)(1) \quad (M1)$$

$$= -0.5 - 1 + 1 \quad \text{A1}$$

$$= -0.5 \quad \text{A1}$$

[6 marks]